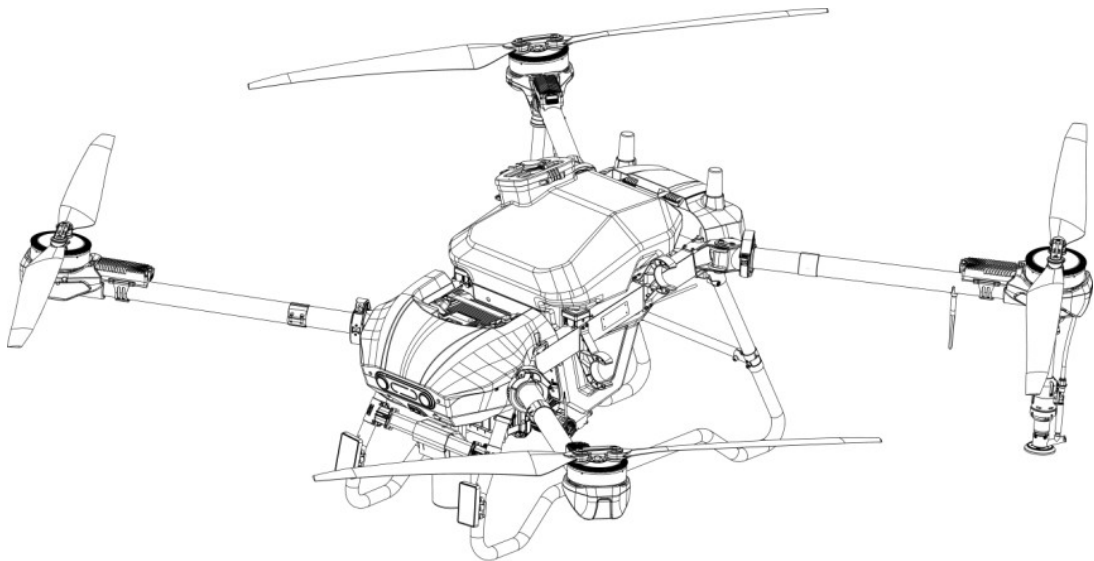


GTEEX REVOLUTION

User Manual V1.3, 2026

INDEPENDENCE SERIES



I-19 Manual

This manual was developed to ensure maximum performance of the INDEPENDENCE 19. Here, you will find detailed information on operation, maintenance and best practices for use, ensuring the best performance and safety in the field. Our goal is to provide an optimized experience for pilots and producers, increasing productivity and application accuracy.

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1. Reading Tips

1.1 Explanation of Symbols

This document uses the following symbols to classify possible risks arising from improper operation:

 "Caution" icons indicate that failure to follow instructions may result in property damage or minor injuries.

 "Warning" icons indicate that failure to follow instructions may result in property damage, serious accidents, or serious injuries.

Read the entire User Manual and familiarize yourself with the product's functions before operating it. Incorrect operation of this product can cause serious injury to you or others, as well as damage to the product and property loss. This product is relatively complex and requires a period of familiarization for safe use, as well as basic knowledge before operation. This product is not suitable for children.

Do not use parts not supplied or recommended by GTEEX REVOLUTION. Users must strictly follow the company's installation and usage instructions. This guide contains safety, operation, and maintenance instructions. Be sure to carefully read all instructions and warnings before assembly, setup, and use.

1.2 Usage Suggestions

GTEEX REVOLUTION provides users with the following documents:

- Item List
- Disclaimer and Safe Operation Guidelines
- User Manual

It is recommended that users use the "Item List" for verification and read the "Disclaimer" and "User Manual" to learn how to use the product. Before use, familiarize yourself with the functions of each component, flight restrictions, emergency return process, and relevant local laws and policies.

If you have any questions during use, please contact GTEEX REVOLUTION or an authorized dealer.

2. Safety Summary

 *The use of INDEPENDENCE 19 may involve certain safety risks and is not suitable for individuals under 18 years of age.*

The safety summary chapter presents only a part of flight safety knowledge. Be sure to read the entire content of the "Disclaimer" and "User Manual" in its entirety carefully.

2.1 Environment

Fly in open fields, away from crowds.

Do not fly above 95ft altitude.

Fly in an environment with temperatures between 32 °F and 104 °F, in good weather conditions (no rain, strong winds, or extreme weather).

Fly only in areas permitted by law. Before flying, consult local aviation authorities to ensure compliance with local laws and regulations and complete the GTEEX REVOLUTION app checklist.

Do not fly indoors.

2.2 Pre-flight Inspection

Ensure all devices have sufficient charge.

Use only genuine GTEEX REVOLUTION parts. If any part is damaged or aged, replace it before flight.

Check that the propellers are not damaged or deformed, are well installed and clean, and correctly installed.

Ensure that the arms and propellers are fully extended and locked. Ensure the motor is clean, intact, and well secured.

Inspect the landing gear, spray tank, and flight battery to ensure they are properly installed.

Ensure that the surfaces of the RTK antenna, image transmission antenna, and radar module are clean and unobstructed.

When prompted by the app, calibrate the compass before flying. Pilots and ground crew must wear helmets and maintain a distance of at least 95ft from the aircraft.

Ensure there are no debris in the operating environment that could interfere with the flight, such as plastic bags, agricultural films, or other light objects.

2.3 Operation

Do not approach the propeller and motor while they are in operation. Maintain a safe distance of more than 50-95ft from the propellers and motor during operation. Ensure that third parties do not enter the operating area throughout the entire process.

Ensure the drone flies without exceeding the recommended maximum weight to avoid dangers.

Do not hang heavy objects on the aircraft and do not modify it in any way, as this may cause flight instability or a crash.

Fly within visual line of sight (VLOS).

Performing an abrupt movement or shutting down the motors in other ways during flight will cause the aircraft to crash. Use this feature only in emergencies.

Critical rotor stop protocol

1. Never stop motors during flight- this will cause immediate uncontrolled descent. Emergency motor cutoff is only permitted when collision with persons is imminent, to minimize injury severity.

2. Emergency stop procedure: Maintain full outward stick deflection for >0.5 seconds to trigger irreversible motor cutoff. The aircraft will enter free fall immediately.



Do not answer phone calls during flight and do not operate the aircraft under the influence of alcohol or drugs.

When there is a low battery warning, immediately return to the safe area.

After landing, first turn off the aircraft, and then the remote control to avoid signal loss and automatic activation of return mode.

Maintain control of the aircraft throughout the entire process and do not rely solely on information provided by the GTEEX REVOLUTION Drone app. Obstacle avoidance will not be available in certain flight modes or environments. Use direct observation to judge the flight situation, avoid obstacles, and set appropriate flight and return altitudes.

If there are high-voltage wires in the operating area, plan the route safely and maintain a safe distance to avoid collisions. If the aircraft hits the wires, do not touch it, as there is a risk of electric shock.

2.4 Flight restrictions and local regulations

Before flying, consult the local air traffic control department to ensure compliance with local laws and regulations. If necessary, obtain authorization from the competent authorities.

Keep the flight altitude below 95 ft and comply with local regulations during flight.

2.5 Protection level description

Under controlled laboratory conditions, the core module of the aircraft has an IP67 protection level. However, this protection may decrease due to aging or wear. Liquid damage is not covered by the warranty.

Situations where protection may fail:

- Collision that deforms the seal.
- Cracks or damage to the sealing coating.
- Protective cover or rubber plug not correctly installed or loose.

2.6 Instructions for pesticide use

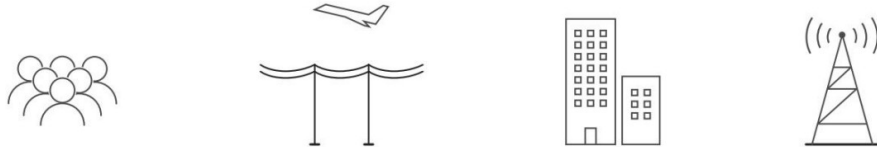
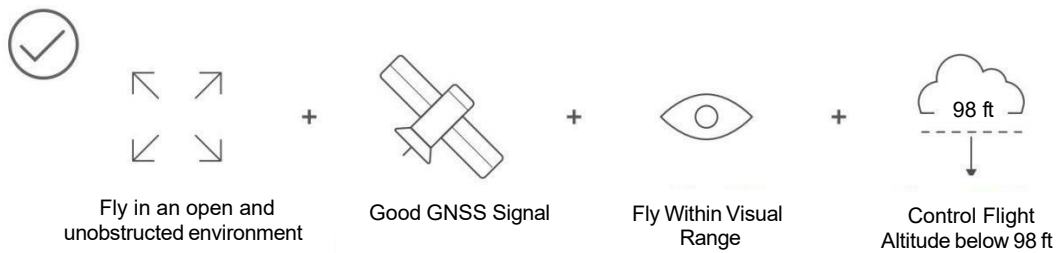
Minimize the use of powdered chemicals and clean them immediately after use to avoid damage to the spraying system. Pesticides are toxic. Use them with caution and follow safety regulations.

During pesticide preparation, be careful with splashes and residues to avoid harm to human health. Use mineral-free water when preparing pesticides. After preparation, filter before adding to the tank to prevent blockages in the spraying system. Ensure operators are downwind to avoid pesticide exposure. Wear protective equipment during application and wash skin, aircraft, and remote control after use.

The effects of pesticides depend on factors such as concentration, flow rate, flight height, wind direction and speed, temperature, and humidity. Consider all these factors to achieve the best results and avoid negative environmental impacts.

When applying pesticides, it is strictly forbidden to pollute rivers and drinking water source. Always follow product panel and precautions.

Disposal of remaining pesticide solution: Plan ahead to reduce the level of remaining pesticide solution to a minimum. The necessary pesticide preparation should be purchased and used according to the plot area. The first step is to spray the remaining pesticide solution and cleaning solution on the crops. Consideration should also be given to installing specialized piping equipment to deal with cleaning solutions.

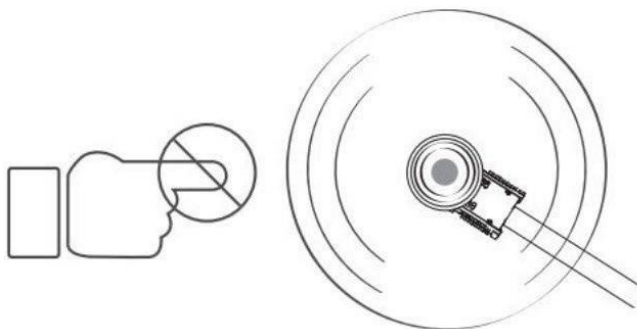


The use of strong acids, strong alkalis, high-temperature liquids, or pesticides prohibited by the government is strictly forbidden.

Please keep away from crowds, power lines, tall buildings, airports, and signal tower lights when flying. Radio transmission towers, high-voltage lines, and substations can interfere with the remote control signal and threaten flight safety. Pay attention to safety during flight.



Do not fly in weather conditions such as heavy rain (precipitation of 1in or more in 12 hours), dense fog, snow, thunder and lightning, or strong winds (wind speed above 14 mph).



Safety distance > 6 ft

Do not touch the rotating propeller. Otherwise, serious personal and property damage may occur.



No-fly zone

3. Product Overview

3.1 Introduction

The INDEPENDENCE 19 agricultural drone, recently developed by GTEEX REVOLUTION, features a spray tank of up to 19 gal and a spreading tank of up to 25 gal (maximum load 175 lbs). It has a stable four-axis architecture and comes equipped with two centrifugal nozzles as standard. The maximum flow rate of 2 centrifugal nozzles can reach 5gal/min, and with 4 centrifugal nozzles, it can reach 8 gal/min. The double-layer spray disc design makes atomization more uniform, and the droplet size is adjustable, which comprehensively improves operational efficiency and quality.

It is equipped with 360-degree rotating radar and terrain-following radar, which can adapt to various types of operating terrains, making flight safer.

The upgraded smart remote control has a 6-inch high-brightness screen and the integrated GTEEX REVOLUTION Drone APP, which further improves the fluidity and stability of operation. The standard high-precision RTK positioning module allows for centimeter-level route planning. It supports both built-in and external battery power, with a total battery life of up to 8 hours, fully meeting the needs of long and intense operations.

Feature Highlights

The INDEPENDENCE 19 agricultural drone adopts a foldable frame structure, achieving the best balance between strength and lightness. It can be quickly folded, facilitating transport and movement.

The onboard high-precision RTK module enables flight routes with centimeter-level positioning and also supports dual-antenna anti-magnetic interference steering technology.

The high-definition FPV front camera monitors the environment ahead in real-time, checks flight conditions, and ensures safety.

The intelligent route operation mode can mark various obstacles within and outside the terrain, perform intelligent takeoff, and fully autonomous operation. It also adds the function of accurate prediction of pesticide breakpoint and performs optimal matching between pesticide level and power, increasing the operational area per unit of power, further automating operations, and improving operational efficiency.

The drone is equipped with 360-degree rotating radar and terrain-following radar, as well as automatic obstacle avoidance and terrain-following flight functions. It can identify common fixed and moving obstacles during the operational process, making operation safer.

Two high-speed, large-flow centrifugal nozzles can achieve an ideal spray width of 50 - 54 ft, according to actual measurements, with a maximum flow rate of 5 gal/min. The droplet size can be adjusted as needed, making it suitable for spraying operations in large fields and high-value crops. It can be expanded to 4 centrifugal nozzles, with a maximum flow rate of 8 gal/min, meeting the needs of ultra-high flow operations.

The INDEPENDENCE 19 agricultural drone is dustproof, splash-proof, and corrosion-resistant. The main components have triple-layer protection, and the overall protection level can reach IP65 (according to IEC 60529 standard of the International Electrotechnical Commission), being resistant to splashes on the fuselage.

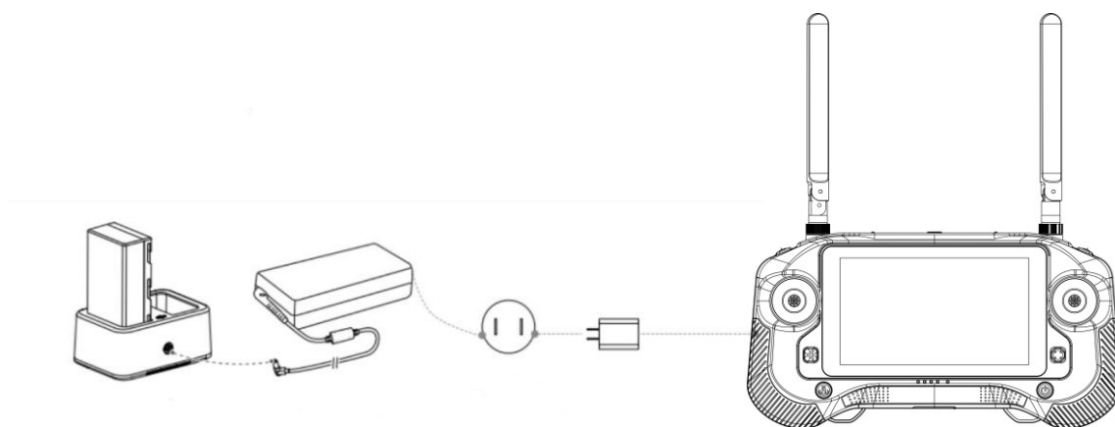
3.2 Prepare the Remote Controller

3.2.1 Powering On and Off

Press and hold the power button for about 2 seconds to turn the remote controller on or off.

3.2.2 Remote Controller Power Supply and Charging

1. **Install the external remote controller battery**
 - a. Press the rear cover opening button to open the cover.
 - b. Insert the external battery into the compartment and push it up to the top.
 - c. Close the cover.
2. **Check battery level:** Users can check the current battery level via the remote controller and the external battery power indicator
 - a. **Check internal battery:** After turning on the remote controller, quickly press the power button once, and the power indicator will display the current battery level.
 - b. **Check external battery:** Quickly press the external battery's power button once, and the external battery's power indicator will display the current battery level.
3. **Charging**
 - a. Use the external battery charging dock and power adapter to charge the external smart battery.
 - b. Use a USB charger and a USB-C cable to charge the remote controller's internal battery.



 To remove the external battery, press and hold the battery release button, then push the battery down to remove it.

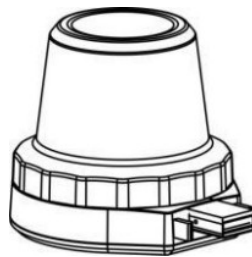
3.2.3 Install the SIM card

Insert the SIM card and wait 30 seconds to check the 4G card detection status. If it shows that the SIM card is inserted, it means the installation is complete. Then, test if the network connection is working smoothly.

 **Attention:** Make sure the SIM card is installed in the correct direction and do not insert it inverted.

3.2.4 Install the RTK marker

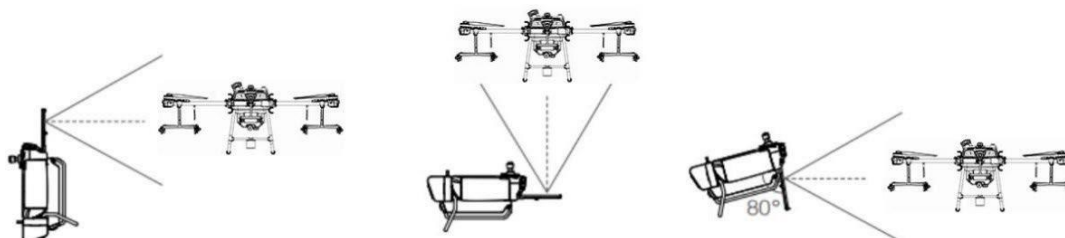
If users plan to use RTK for planning the operating area, they will need to insert the RTK marker into the USB interface of the remote control.



3.2.5 Adjust the Antenna

Unfold the remote controller antenna and adjust its position. Different antenna positions receive different signal strength levels. When the angle between the antenna and the back of the remote controller is 80° or 180° , and the antenna plane faces the aircraft, the signal quality between the remote controller and the aircraft can be optimized.

During aircraft operation, ensure it is kept within the best communication range. Adjust the orientation or distance between the operator and the aircraft as needed, ensuring the aircraft is always within the optimal communication range.



 *Do not use other communication devices on the same frequency band simultaneously to avoid interference with the remote control signal.*

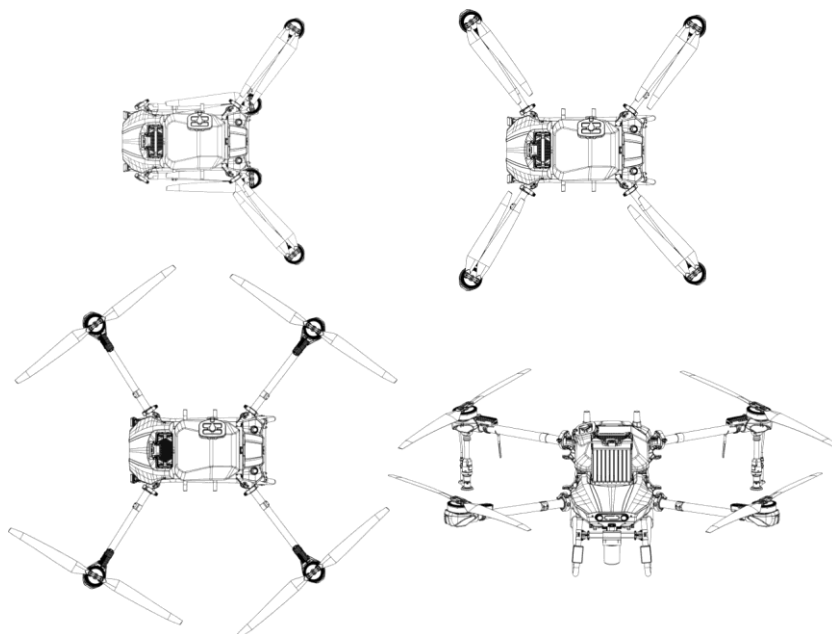
If users use the RTK marker for RTK planning, it is necessary to remove the module after planning is complete, otherwise, the communication performance of the remote control may be affected.

3.3 Prepare the Aircraft

Unfold arms M4 and M3 first, then arms M1 and M2, in that order. After unfolding all arms, ensure that all 4 arm locks are firmly secured.

When locking the aircraft arm, users may feel relatively greater resistance. Press firmly to lock the latch.

Unfold the propellers individually.



Gently insert the battery into the compartment until you hear a click indicating the battery is in place.

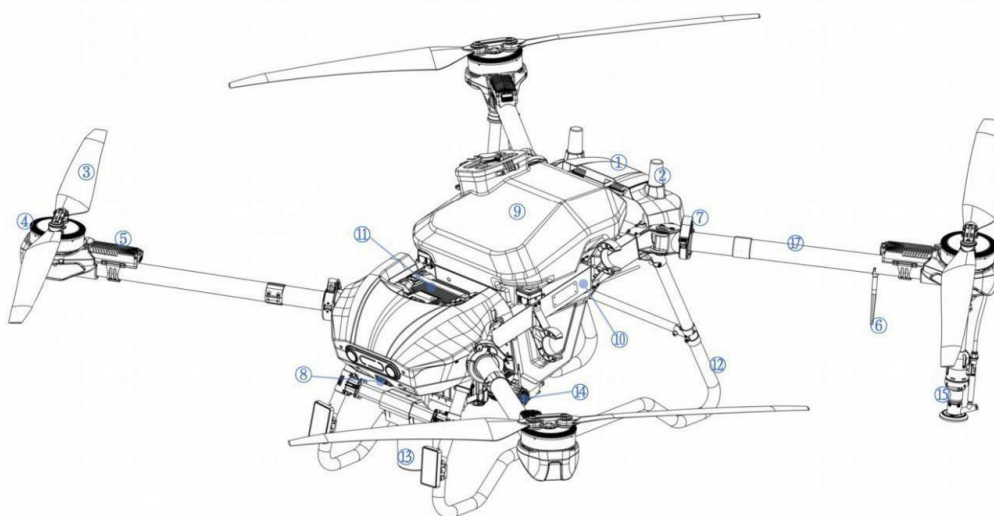
 *Ensure the battery is properly installed. Check that the battery is off when removing or inserting it.*

To remove the battery, press the battery retention latch, then pull it up.

If you need to fold the arms again, follow the reverse order: fold arms M1 and M2 first, then arms M3 and M4. Make sure the arms are secured in the storage clips located on the sides of the aircraft, to avoid damage to the arms.

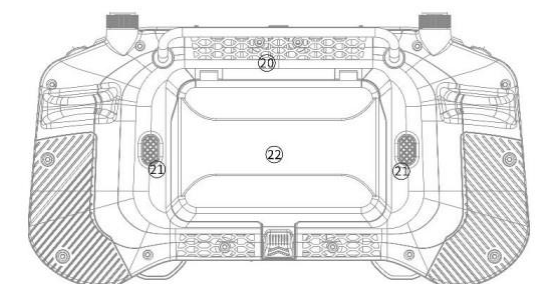
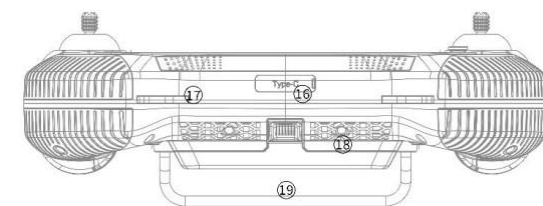
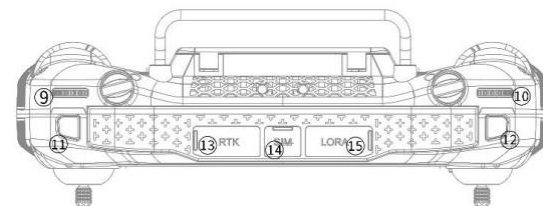
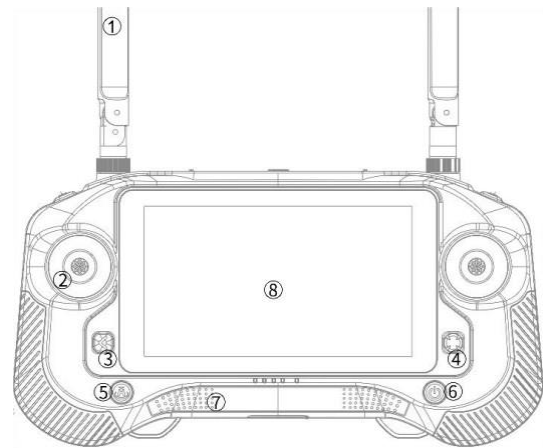
3.4 Aircraft Components

1	Avionics module	2	RTK antenna
3	Propeller	4	Motor
5	ESC (Electronic Speed Controller)	6	Transmission antenna
7	Arm lock	8	Front FPV camera
9	Spray tank	10	Aircraft frame
11	Smart battery	12	Tripod
13	Radar	14	Impeller pump
15	Centrifugal nozzle	16	Arm carbon tube
17	Navigation lights		



3.5 Remote Control Components

1	Antena
2	Joystick
3	Back button
4	Multi-directional button (five directions)
5	Return button
6	Switch
7	Speaker sound output
8	Touch screen
9	Reserved rotary button
10	Rotary button for operational flow adjustment
11	Radar switch
12	Spray switch
13	External RTK interface
14	SIM card slot
15	LORA interface
16	TYPE-C interface
17	Lanyard hole
18	Vents (ventilation)
19	Handle
20	Air outlet
21	Customizable buttons
22	Battery cover



3.6 Activation

A new aircraft needs to be activated. Follow the steps below to complete the activation:

- Turn on the remote control, then turn on the aircraft.
- Wait for the aircraft to complete self-check and automatically connect to the remote control. The remote control status indicator will turn green, indicating a successful connection to the aircraft.
- Open the GTEEX REVOLUTION Drone APP, enter your username and password already provided by the dealer, and it will be available for flight.

4. Flight Safety

4.1 Requirements for the Operating Environment

To avoid personal injury and property damage and ensure the effectiveness of operations:

Wind speed: Operate in an environment with a wind speed below 14 mph. For herbicides, fungicides, and insecticides prone to drift or toxicity, it is recommended to operate in an environment with a wind speed below 7 mph.

Weather conditions: Do not fly in adverse weather conditions, such as strong winds (14 mph or more), rain (1 in or more in 12 hours), snow, fog, etc.

Flight location: Choose an open area without tall buildings around. Tall buildings can block the GNSS signal, impairing or preventing aircraft localization.

Obstacles: During takeoff, landing, and flight, pay attention to obstacles such as power poles, high-voltage lines, and trees. Keep away from roads, bodies of water, people, and animals.

Line of sight: Always fly within visual range and away from obstacles, people, animals, water, etc.

Electromagnetic interference: Avoid areas with electromagnetic interference, such as high-voltage lines, communication base stations, or transmission towers, near or within the operating area.

Maximum altitude: Do not fly above 300 ft altitude. As altitude increases, the aircraft's load capacity decreases.

The **GTEEX REVOLUTION** application will display the payload weight in the spray tank. The maximum weight of material loaded by the user must not exceed the recommended value, otherwise, it may compromise flight safety. Ensure that the GNSS signal is good and the RTK antenna is not blocked during operation. **Do not operate the aircraft indoors.**

4.2 Flight Limit

GTEEX REVOLUTION has always prioritized flight safety. For this purpose, it offers a variety of technical means to help users operate the aircraft in compliance with local regulations. It is strongly recommended that users download and install the latest firmware in the assistant software to ensure the proper functioning of the flight restriction feature.

The flight restriction function includes:

4.2.1 No-fly zone

No-fly zones include, but are not limited to, large airports, major cities/regions, and temporary event areas worldwide.

Different levels of flight restrictions will be applied in different no-fly zones, including, but not limited to:

- Warnings
- Prohibited takeoffs
- Flight altitude restrictions
- Forced landings in no-fly zones

All flight-related functions will be affected to different degrees when the aircraft is near or within a no-fly zone. This includes:

- Reduction in aircraft speed
- Inability to set up flight missions
- Interruption of ongoing flight missions

4.2.2 Height and Distance Limits

1. **Height Limit:** Keep the flight altitude below 95 ft and keep the aircraft away from tall buildings during flight.
2. **Distance Limit:** The standard maximum flight distance is 3280 ft, and can be adjusted up to 6562 ft. If the flight distance is too long, the remaining battery power may not be sufficient for the aircraft to return, so fly with caution.

4.3 Pre-flight Inspection

Ensure all devices have sufficient power. Use only original GTEEX REVOLUTION parts and check that all parts are in good condition. Replace any worn or damaged parts before flight. Ensure that the propellers are not damaged or deformed, are installed correctly, and that the surface is clean and free of debris.

Check that the arms and propellers are fully extended and that the arm locks are firm. Ensure that the motor is clean, undamaged, well installed, and functioning normally. Check that the landing gear, spray tank, and flight battery are securely installed. Ensure that the surfaces of the RTK antenna, image transmission antenna, and radar module are clean and unobstructed.

When the application prompts for compass calibration, perform the calibration before flying. Pilots and crew members should wear helmets and maintain a safe distance of more than 30m from the aircraft. Ensure that the operating environment is free of debris that could affect flight, such as plastic bags, empty fertilizer packaging, plastic films, and other objects easily blown by the wind. Ensure that the GTEEX REVOLUTION **Drone APP** is running to assist with flight.

4.4 Calibrate Weighing Scale and Flowmeter

4.4.1 Calibrate the weighing scale

1. **Situations requiring scale calibration**
 - a. First use.
 - b. After replacing the seed tank or spray tank.

- c. When the difference between the actual operating area and the theoretical area is greater than 10%.
- 2. **Calibration steps**
 - a. Empty the spray tank or seed tank.
 - b. Access the app's operation interface, click **Settings** > **Spray Settings/Spreading Settings**, and click the **Calibrate** button to the right of the weighing scale calibration option.
 - c. Click **Start Calibration**, and the system will automatically perform the calibration.
 - d. Wait approximately 5 seconds for the calibration results to be displayed.
 - e. If calibration fails, the system will indicate the reason for failure. Follow the instructions in the GTEEX REVOLUTION INDEPENDENCE 19 User Manual to resolve the problem, then recalibrate and wait until calibration is successful.

4.4.2 Calibrate the flowmeter

- 1. **Situations requiring flowmeter calibration**
 - a. First use.
 - b. Replacement of the solution with another of different viscosity.
 - c. When the difference between the actual operating area and the theoretical area is greater than 15%.
- 2. **Calibration steps**
 - a. Add about 5 gal of water to the spray tank.
 - b. Access the app interface, click **Settings** > **Spray Settings**, and click the **Calibrate** button next to the flowmeter calibration.
 - c. Click **Start Calibration**, and the system will automatically perform the calibration and display the results.
 - d. If calibration is successful, normal spraying operations can be performed.
 - e. If calibration fails, the system will show the reason for the error. Follow the instructions to correct the problem and try again.

4.5 Calibrate IMU and Compass

- 1. **Situations requiring IMU or compass calibration**
 - a. First use.
 - b. When the app indicates that the IMU or compass needs to be calibrated.
 - c. It is recommended to calibrate the IMU and compass after long-distance transport exceeding 125 miles.
- 2. **IMU calibration steps**
 - a. Access the app interface, click **Settings** > **Aircraft Settings**, scroll to the end, select **Advanced Settings** and then **IMU and Compass Calibration**.
 - b. Place the aircraft horizontally and stationary with arms folded, click the **IMU Calibration** button in the app, and the system will automatically perform the calibration and display the results.
 - c. If calibration fails, the system will show the reason for the error. Follow the instructions to correct the problem and try again.
- 3. **Compass calibration steps**

- a. Access the app interface, click **Settings** > **Aircraft Settings**, scroll to the end, select **Advanced Settings**, and click **Compass Calibration**.
- b. Lift the aircraft approximately 4 feet from the ground and rotate it 2-3 times horizontally counter-clockwise. If the app indicates that calibration was successful, it is complete.
- c. If calibration fails, tilt the aircraft slightly and repeat the procedure.
- d. If it still fails, restart the calibration process.

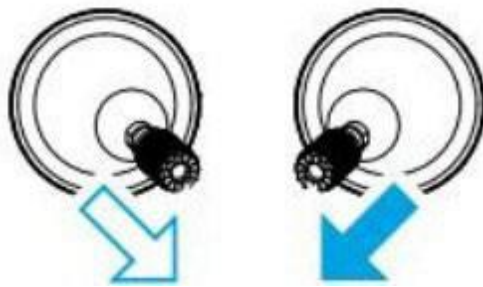
4.6 Powering On/Off the Motor

1. Powering on the motor

- a. Execute the **Combined Sticks Command (CSC)** and hold for 2 seconds to power on the motor.
- b. After the motor powers on, immediately release the joystick and take off as quickly as possible.
- c. If you cannot take off, do not attempt to start the motor again with the joystick, as this may cause instability, drift, or even accidental takeoff, resulting in property damage or personal injury.

2. Powering off the motor

- a. Pull the throttle stick down until the aircraft lands, holding it in the lowest position for 3 seconds to power off the motor.



⚠ *High-speed propellers are dangerous. Maintain a safe distance from the aircraft and from people, animals, or obstacles. Before powering off the motor, ensure you have the remote control in your hands and that the aircraft is fully under control. Do not power off the motors during flight, unless it is an emergency (e.g., to avoid a collision with people). After landing, first power off the aircraft, then the remote control.*

4.7 Basic Flight

4.7.1 Takeoff

1. Place the drone in a visible and safe location in the operating area, with the tail facing people.
2. Add the liquid pesticide or spreading material to the spray tank and close the lid tightly.
3. Turn on the remote controller and ensure the ****GTEEX REVOLUTION Drone APP**** is functioning normally. Then, turn on the aircraft and check that it is connected to the remote controller.

4. Users should wear helmets and maintain a safe distance of more than 30 meters from the aircraft.
5. If using RTK positioning, ensure the RTK function is activated and select the correct signal source (mobile RTK base or network RTK service).
6. Wait for satellite search to ensure the GNSS signal is good and RTK is functioning normally.
7. After RTK positioning is fixed, move the internal sticks to start the motor.
8. Push the throttle stick up immediately after the motor starts to make the aircraft take off smoothly.

4.7.2 Safe Parking

1. To descend, ensure the operation has ended. Manually control the aircraft, slowly pulling the throttle stick down, so that it descends smoothly to a flat surface.
2. After landing, keep the stick in the lowest position for more than 3 seconds until the motor completely stops.
3. Ensure the motors are off before powering off the aircraft and the remote control.

 *During flight, if the app indicates that the aircraft battery is in a low charge state, users should take the aircraft to a safe area and land as soon as possible, then replace the battery. If the app indicates that the aircraft battery is in a severely low charge state, the aircraft will automatically land on the spot. During the aircraft's automatic landing, if users need to use the stick to adjust the aircraft's position, make sure to do so cautiously.*

5. Aircraft

5.1 Flight Modes

The aircraft has two flight modes: normal and attitude. Normal mode is used by default. If attitude mode is allowed in the app, it can be changed via the flight mode switch on the remote control.

Normal Mode (P): Allows for precise hovering and aircraft positioning, and centimeter-level positioning can be achieved when using RTK.

Attitude Mode (A): Fixed hovering cannot be performed, only attitude stabilization is provided. The aircraft's flight speed in attitude mode depends on environmental factors, such as wind speed.

Notes on Attitude Mode

In attitude mode, the aircraft is easily affected by external factors, which can cause horizontal drift. Therefore, the aircraft cannot hover at a fixed point in this mode, and the user needs to manually control the remote control to achieve hovering.

In this mode, the aircraft will not be able to locate itself, and the control difficulty will be much higher. If users use this mode, they must be familiar with the aircraft's behavior in this mode and be able to control it skillfully. When using it, avoid flying long distances to avoid the risk of losing judgment of the aircraft's attitude due to distance.

At the same time, users should avoid flying in environments with weak GNSS signals, obstructions (such as near tall buildings), and narrow spaces to avoid passively entering attitude mode and causing flight accidents.

5.2 Operation Modes

Operation modes include route operation, AB operation, manual operation, and enhanced manual operation. Users can select the corresponding operation mode for spraying or spreading according to different operation scenarios. See more details in chapter 7.4 on operation mode.

5.3 Operation Recovery

If users exit route operation or AB operation in the middle of the process, the aircraft will record the breakpoint, and the user can continue the operation through the operation recovery function. The operation recovery function is mainly used to refill the spray tank, replace the aircraft battery, or avoid obstacles during operation.

5.3.1 Recording breakpoints

During route operation or AB operation, if the GNSS signal is good, the following operations will cause the aircraft to record a breakpoint. If the GNSS signal is weak, the aircraft will

enter attitude mode, exit the current operation, and record the position when the last good GNSS signal was received as the breakpoint.

Click the "Pause" button in the lower right corner of the application; The aircraft enters the return process in any way; The remote control's tilt or rotation stick is moved; When an obstacle is detected, the aircraft brakes suddenly and enters obstacle avoidance mode;

The obstacle avoidance function of the radar module is activated, but an anomaly is detected and the radar cannot operate normally; The aircraft has reached the flight limit or is about to enter a restricted flight zone; The spray tank is out of pesticide; The aircraft triggers the low battery alarm;

When the connection between the aircraft and the remote control is disconnected, if the "continue operation after connection loss" option is not activated in the aircraft settings, the breakpoint will be recorded when the aircraft performs the action after connection loss;

When something goes wrong with the spraying/spreading equipment.

 *Ensure a good GNSS signal when using this function, otherwise, breakpoints may not be accurately recorded or returned to. The system will update the breakpoint whenever any of the above conditions are met.*

5.3.2 Continue operation

Exit the operation by any of the methods above, and the aircraft will record the breakpoint (i.e., the break point).

After the aircraft performs the necessary operations (such as replacing the battery, refilling the pesticide, or controlling the aircraft to move it away from obstacles, etc.), control the aircraft to the appropriate position.

To select the aircraft's return route, the user can select a breakpoint on the right side of the operation interface or select a return point from the display list.

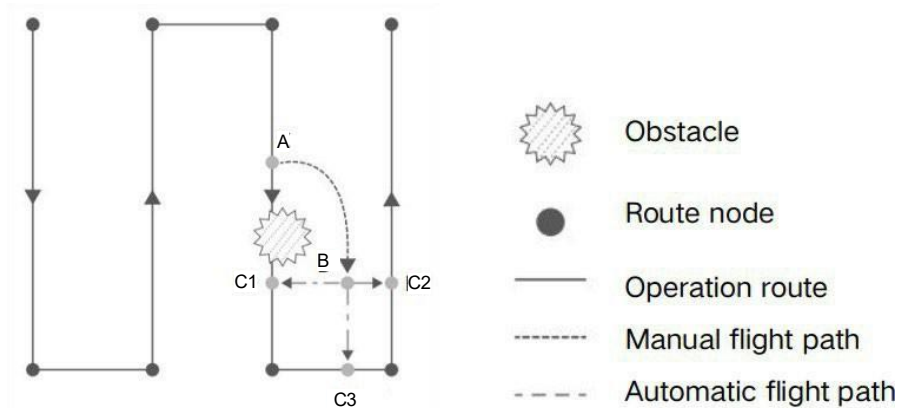
Click the "Continue" button in the lower right corner of the APP, and the aircraft will fly to the selected breakpoint or return point, and then continue the operation. The intelligent takeoff function is supported when returning. The aircraft will return to the breakpoint via the added transfer point or automatically bypass added obstacles.

 *If a transfer point is added before continuing the operation, click "Continue" and the aircraft will pass through the transfer point and reach the breakpoint after takeoff. If users call the "in progress" route, the transfer points added before the operation will become invalid. If users want to avoid obstacles, they can add transfer points again before continuing the operation.*

After activating the obstacle avoidance function, the aircraft will detect obstacles when returning to the breakpoint or return point. If the aircraft triggers obstacle avoidance hover, the user will need to manually control the aircraft's direction using the stick. See more details on "manual obstacle avoidance" below.

5.3.3 Manual obstacle avoidance

During route operations and AB operations, if the aircraft triggers obstacle avoidance hover, or other emergency situations occur (such as abnormal aircraft behavior), the user can control the aircraft's flight direction via the remote control to manually avoid obstacles or deal with emergency situations.



1. Exit route operation or AB operation

During operation, if there are obstacles on the operation route that need to be avoided, control the aircraft's left and right direction via the remote control (the remote control has a rotation stick and stick action), the aircraft will automatically exit the operation and record breakpoint A, starting to switch to manual operation mode.

2. Bypass obstacles

After switching to manual operation mode, the user can control the aircraft to bypass obstacles via the remote control and fly from point A to point B when exiting the operation.

3. Continue operation

In the list of breakpoints and return points displayed in the APP, select one of the return points C1, C2, and C3, and click the "Continue" button. The aircraft will fly from point B to the selected return point along the path perpendicular to the operation route, and then continue the operation.

 During the process of continuing the operation and returning to the original operation route, if an emergency occurs (such as obstacles on the return route, etc.), the above operations can be repeated.

 The number and location of optional return points are related to the aircraft's position and operation mode. For example, during route operations, there is no C3 return point on the non-spraying route. Make your selection according to the APP prompts.

Ensure the safety of the aircraft, make sure the aircraft has completed bypassing obstacles before continuing the operation, otherwise, it will be dangerous. When dealing with other emergencies, ensure the aircraft returns to normal, then manually fly to the correct position before continuing the operation.

5.4 Tank Empty Point

The aircraft will calculate the empty tank point based on the user-defined remaining application area, the current amount of volume in the spray tank, the current aircraft status, and operating parameters, and display it on the map. The user can configure the aircraft's behavior to hover or return after spraying is completed in the application. After adding materials to the spray tank or seed tank, this function is automatically activated and the "empty tank" icon will be displayed on the map in real-time.

5.5 Safety System

The aircraft is equipped with terrain-following radar and obstacle avoidance radar, which can accurately detect the relative height of crops and perform terrain-following without aerial measurements. It can also detect obstacles in various directions and perform autonomous obstacle avoidance after detecting them during operation. During landing, the descent speed can be limited according to the aircraft's height relative to the ground as measured by the radar to ensure a smooth landing.

5.5.1 Obstacle Avoidance Function

In the app's operation interface, click Settings \> Radar Settings in the upper right corner to enter the radar settings interface and enable the obstacle avoidance switch. Users can also enable obstacle avoidance by pressing the button on the remote control. Once enabled, the aircraft will enter obstacle avoidance state upon detecting an obstacle.

The user can use the control to make the aircraft fly away from the obstacle, as instructed in the app. It can be used in the following two situations:

- When the distance to the obstacle is large, the aircraft will begin to decelerate upon detecting the obstacle until it hovers. During the deceleration process, the user can control the aircraft to fly away from the obstacle via the remote control.
- When the aircraft is close to an obstacle, it will suddenly brake and hover. After hovering, the aircraft will be in the obstacle avoidance state. The app will inform the direction to move the control stick. The user can move the stick in the opposite direction of the obstacle, according to the warning, and exit the obstacle avoidance state to regain control.

 *Obstacle detection may fail for small obstacles, diagonal wires, and objects aligned with the aircraft's tripod. Please fly with caution. Manually control the aircraft when necessary to avoid flight accidents.*

5.5.2 Terrain Following Function

In the app's operation interface, click Settings \> Radar Settings in the upper right corner to enter the radar settings interface, select the operation scenario as "Flat Terrain", "Mountains or Fruit Trees", or "Water Surface", and activate the terrain following and deviation functions, respectively. After starting the operation, the aircraft will perform terrain-following flight

according to the configured relative crop height and will deviate to avoid obstacles after detecting them.

1. Flat Terrain

This operation scenario is suitable for field crops with small height variations and fruit tree crops with small undulations at the top. The aircraft will prioritize deviating from obstacles laterally after detecting them. Automatic deviation can be interrupted by moving the control during deviation. If the aircraft cannot deviate, it will hover in place, and the user can manually deviate from the obstacle by moving the control.

2. Mountains or Fruit Trees

This scenario is suitable for crops in mountainous terrain with large height variations and fruit tree crops with large undulations. The aircraft will prioritize deviating from obstacles from top to bottom after detecting them. Automatic deviation can be interrupted by moving the control during deviation. If the aircraft cannot deviate, it will hover in place, and the user can manually deviate from the obstacle by moving the control.

3. Water Surface

This scenario is suitable for operations in ponds and rice fields where crops are submerged in water. The aircraft will prioritize deviating from obstacles laterally after detecting them. When the terrain following function is activated, the aircraft will use the radar height relative to the ground as the fixed flight altitude.

 *Choose the operation scenario according to the actual terrain, otherwise, this may lead to inaccurate terrain following and failure in obstacle avoidance, such as: when the "Mountain or Fruit Trees" scenario is selected for operations on flat terrain, the aircraft may experience abnormal ascent in some scenarios (power poles, trees).*

When the "Flat Terrain" or "Water Surface" scenario is selected for operations in mountainous terrain, the aircraft will deviate laterally first, which may cause the deviation to fail and affect the spraying effect.

Manual operation mode only supports the obstacle avoidance function while the aircraft hovers. In some scenarios (such as power lines and small obstacles), the obstacle avoidance function may be affected, and users can manually control the aircraft to deviate.

When the aircraft is on the water surface, the terrain following function may be affected, so fly with caution. Remember that the operating height must be greater than 2m, otherwise, the aircraft may crash.

5.5.3 Precautions for Radar Use

When the aircraft is powered on or after flight, do not directly touch the metal part of the radar module with your hands, as this may cause burns.

Maintain control of the aircraft throughout the entire process and do not rely entirely on the information provided by the radar module and the GTEEX REVOLUTION app. Ensure good control, use visual observation, and reasonably judge flight conditions to avoid obstacles in time.

In manual operation mode, since the user can freely control the aircraft, pay attention to the flight speed and direction during operation to prevent obstacles from entering the radar's detection blind spot and causing collisions. Use the obstacle avoidance function appropriately according to the surrounding environment.

The obstacle avoidance function cannot be used in attitude mode.

For objects such as slanted cables, power poles with a significant incline (above 10°), and cables that are at an angle relative to the aircraft's flight direction, most of the radar's electromagnetic waves will be emitted in other directions, which will affect the radar's detection performance. Fly with caution.

If the distance between the aircraft and the measured surface exceeds the radar module's altitude range, the terrain following function may fail. Always pay attention to the relative altitude between the aircraft and the measured surface.

Use the radar module with caution over surfaces of objects whose inclination exceeds the following values: 10° (aircraft speed must be less than 3ft/s), 6° (aircraft speed must be less than 10ft/s), 3° (aircraft speed must be less than 16ft/s).

For vertical objects, if the lower radar is higher than the top of the object, most of the radar's electromagnetic waves will be reflected in other directions, which will affect the radar's detection performance, so fly with caution.

Use the radar module according to local radio regulations and applicable laws. Radar sensitivity may be reduced when multiple aircraft are operating in nearby areas, so fly with caution.

The radar module is relatively precise, so do not squeeze or strike it.

Before each use, check to ensure that the appearance is clean, the outer protective cover is intact and without cracks, dents, or other deformities.

5.6 Automatic Return

The aircraft has functions such as intelligent return, low battery return, and loss of control return.

- **Home point:** When armed and with a good GNSS signal, the aircraft's current position will be recorded as the origin point.

- **Return:** The process of the aircraft automatically returning to the origin point is called return.

5.6.1 Smart RTH

Smart RTH can be activated by pressing the RTH button on the remote control. The process is the same as fail-safe RTH, except that the user can control the aircraft's altitude to avoid obstacles using the sticks. After pushing the pitch or roll stick to exit Smart RTH, the user automatically regains control.

5.6.2 Low Battery Return

If the behavior after reaching low battery is configured to return to the origin point in the smart battery settings in the APP, when the aircraft battery reaches the low battery limit during flight, it will automatically suspend the operation and enter the return process. During the return process, the user can use the stick to control the aircraft's height and avoid obstacles. After pushing the pitch or roll stick to exit smart return, the user automatically regains control.

If the behavior after reaching low battery is configured to alarm in the smart battery settings in the APP, the aircraft will not enter low battery return.

5.6.3 Return after Loss of Control

Users can configure the aircraft's behavior after remote control signal loss in the APP to return to the origin point, hover, or land. If configured to return to the origin point, when the aircraft successfully records the origin point and the remote control signal is lost for more than 3 seconds, the flight control system will cause the aircraft to return to the most recently recorded origin point.

If the remote control signal returns to normal during the return process, the return process will continue, but the user can control the flight via the remote control and can cancel the return process by pressing the stick and regaining control.

5.6.4 Safety Precautions for Automatic Return

- **GNSS Signal:** When the GNSS signal is weak (red GNSS icon) or GNSS is not operating, automatic return cannot be used.
- **Return and Obstacle Avoidance:** When the environment meets the radar module's operating conditions, the aircraft can automatically return to the origin point and avoid obstacles.

 *If there are obstacles on the return path, the aircraft will automatically deviate or decelerate until it hovers. After hovering, the aircraft will exit the return process and the user can manually control the aircraft. If the return is performed during route operations, the aircraft will automatically plan a path to bypass obstacles added when returning.*

If a transfer point is added before performing the operation, the aircraft will pass through the transfer point and return to the origin point upon return, improving the safety of the return

path. Once the operation has started, transfer points cannot be deleted, and transfer points must be replanned after the operation is completed.

If it is not necessary to pass through a transfer point to return to the origin point, users should exit automatic return and manually control the return.

- **Landing Protection Function:** During the automatic return process, when the aircraft reaches the origin point, it will activate the landing protection function and hover. The user can control the aircraft to a suitable ground location via the remote control to ensure a safe landing.

 *When using RTK positioning and RTK is functioning normally, the aircraft will land directly on the ground and will not enter the landing protection process. When returning home out of control, the aircraft will land directly on the ground and will not enter the landing protection function.*

5.7 Low Power and Low Voltage Protection

The aircraft has low battery alarm, very low battery alarm, and low voltage alarm functions.

If the APP displays a low battery alarm, fly the aircraft to a safe area and land as soon as possible, then replace the battery. If the behavior after reaching low battery is set to return, the aircraft will automatically return after the low battery alarm appears in the APP. If there is a very low battery alarm or low voltage alarm in the APP, the aircraft will automatically land on the spot. The user cannot cancel the landing, but can control the horizontal direction of the aircraft via the remote control.

 *The low battery limit can be set in the APP.*

5.8 RTK Function

The aircraft comes with onboard RTK. Compared to the compass module, the onboard dual-antenna steering technology not only offers higher accuracy but also provides strong resistance to electromagnetic interference, ensuring reliable operational flight in environments with strong magnetic interference, such as high-voltage lines and metallic buildings. Dual-antenna steering will be automatically activated when the GNSS signal is good.

The aircraft cooperates with mobile base stations or RTK services on the network to achieve centimeter-level positioning, improving the accuracy of agricultural protection. The specific use is as follows

5.8.1 Enable/disable RTK function

Before using RTK positioning, check if the RTK signal reception method (mobile base station or RTK service on the network) is correctly selected. Otherwise, RTK positioning cannot be used. Go to the APP operation interface \> Settings \> RTK Settings to view and configure.

If users do not use RTK data, uncheck the "RTK Detection" option, otherwise, the aircraft will not be able to take off without RTK data.

5.8.2 Use with mobile base stations

Refer to the relevant equipment documentation to complete the frequency matching between the aircraft and the base station, as well as the base station installation.

Turn on the base station and wait for satellite search. When the RTK status icon above the APP operation interface is green, it means the aircraft has acquired and is using the RTK data from the base station.

5.8.3 Use with network RTK service

The network RTK service uses the remote control instead of the base station to connect to the designated network RTK server to send and receive RTK data. Keep the remote control on and connected to the internet during use.

Ensure that the remote control shows that the device is connected to the internet.

Go to the APP operation interface \> Settings \> RTK Settings, and select the RTK signal source as "Network RTK".

Wait until the connection to the network RTK server is established. When the RTK status icon above the operation interface turns green, it means that the network data has been acquired and is being used.

5.8.4 Custom Network RTK Service

When using a third-party network RTK service provider, follow the instructions below to configure it.

Ensure that the remote control shows that the device is connected to the internet.

Go to the app interface \> Settings \> RTK Settings and select the RTK signal source as "Custom network RTK".

When using for the first time, click "Edit", enter the server-related parameters, and click "OK" when done.

Wait for the connection to the server to be established. When the RTK status icon on the operation interface turns green, it means that the network RTK data has been obtained and is being used.

5.9 Flight Indicator Light and Night Light

5.9.1 Indicator Light

The aircraft's M1 to M4 arms are equipped with LED lights.

The LED lights on arms M1 and M2 are the aircraft's front indicator lights. During flight, the red light is continuously on to indicate the aircraft's front direction.

The LED lights on arms M3 and M4 are the aircraft's status indicator lights, which flash to represent the aircraft's equipment status.

When the aircraft is not taking off, all indicator lights are off. When the aircraft is on the ground and the motors are running, the LED lights begin to light up and flash as described below.

LED light definitions for M3 and M4	
Successful unlock, successful lock	Green light on for 1 second
Unlock failed	Red light on for 1 second
Intense IMU vibration, abnormal data, IMU not calibrated	Yellow light flashes slowly
Remote control signal loss	Yellow light flashes slowly
No GNSS	Red light flashes slowly
With GNSS	Green light flashes slowly
After activating RTK function, RTK not located	Yellow and green lights flash alternately
Abnormal compass data and compass not calibrated	Red lights flash alternately
IMU leveling	Yellow light on
IMU calibration successful	Green light on for 1 second
IMU calibration failed	Red light on for 1 second
Compass leveling	Green light on
Compass calibration successful	Green light on for 1 second
Compass calibration failed	Red light on for 1 second
Remote control pairing	Yellow light on
Remote control pairing successful	Green light on for 1 second
Remote control pairing failed	Red light on for 1 second

5.9.2 Night light

Users can choose to equip the aircraft with night lights to improve operational safety at night. Access the GTEEX REVOLUTION Drone app operation interface and enable or disable the night lights in the settings.

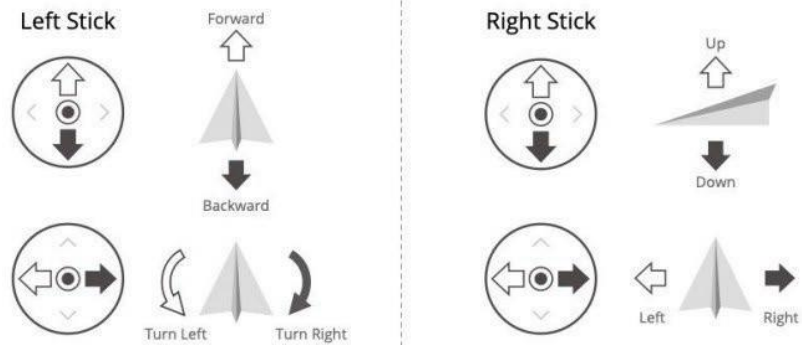
 *Do not directly illuminate eyes with the night lights to avoid potential eye damage due to intense light.*

6. Remote Control

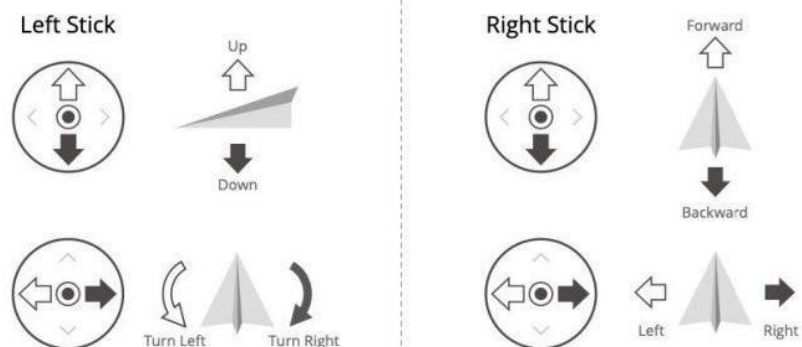
6.1 Remote Control Joystick Mode

Before using the Remote Control stick to control the aircraft, users need to clarify the current stick mode of the aircraft to ensure flight safety. The Remote Control stick modes are divided into **Mode 1**, **Mode 2**, and **Mode 3**, as shown in the figure below.

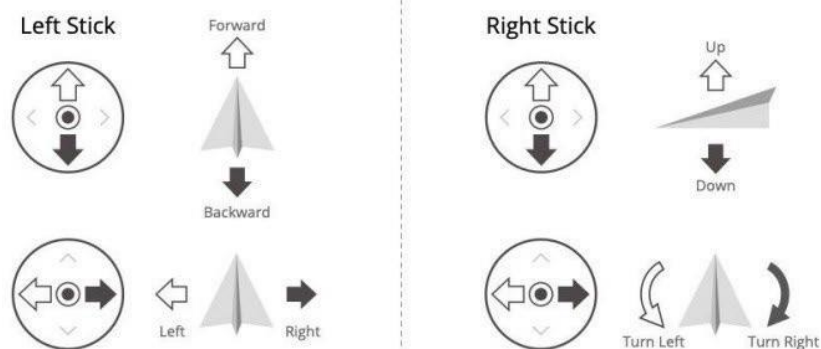
Mode 1



Mode 2

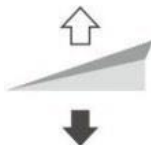
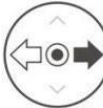
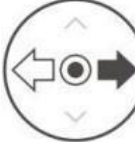


Mode 3



 **Joystick return to center/neutral position:** The remote control joystick is in the intermediate position. **Joystick amount:** The amount by which the remote control joystick deviates from the joystick's center position.

This manual uses **Mode 1** as an example to explain how to control the Remote Control.

Left Joystick 		The throttle stick is used to control the aircraft's ascent and descent. Push the stick up to raise the aircraft. Pull the stick down to lower the aircraft. The aircraft's altitude remains constant in the center position (automatic altitude setting). When the aircraft takes off, the throttle stick must be pushed up past the center position for the aircraft to take off from the ground.
Left Joystick 		The yaw stick is used to control the aircraft's direction. If the user pushes the stick to the left, the aircraft will rotate counter-clockwise. If the user pushes the stick to the right, the aircraft will rotate clockwise. The rotation angle is zero when the stick is in the neutral position, and the aircraft does not rotate.
Right Joystick 		The pitch stick is used to control the aircraft's forward and backward flight. Push the stick up, and the aircraft will tilt forward and fly forward. Pull the stick down, and the aircraft will tilt backward and fly backward. In the neutral position, the aircraft's front and rear directions remain level.
Right Joystick 		The roll stick is used to control the aircraft's lateral flight, to the left and right. Move the stick to the left, and the aircraft will tilt left and fly in that direction. Move the stick to the right, and the aircraft will tilt right and fly in that direction. In the neutral position, the aircraft's left and right directions remain level.

6.2 Remote Control Joystick Calibration

The remote control has already been calibrated before leaving the factory. During daily use, if the joystick does not return to the center or has a significant deviation, users can calibrate the joystick.

Calibration process:

- Access the app's operation interface \> Settings \> Remote Control Settings and click the "Calibrate" button.
- Follow the app's instructions and move the left and right joysticks to their maximum limits.
- After the app indicates that calibration was successful, calibration is complete.

6.3 Remote Control Pairing

At the time of factory departure, the remote control has already been paired with the aircraft's internal receiver. After powering on, it will be ready for use. If it is necessary to replace the remote control, a new pairing must be performed before use.

Pairing process:

- Turn on the remote control, open the GTEEX REVOLUTION Drone app, and then turn on the aircraft.
- Access the app's operation interface **Settings \> Remote Control Settings** and click the **Pair** button.
- Lift arm 4 of the aircraft and tilt it forward by more than 20°.
- If pairing is successful, the app will display the message "Aircraft connected". If pairing fails, it will be necessary to return to the pairing state and repeat the process.

7. GTEEX REVOLUTION Drone App

The GTEEX REVOLUTION Drone application was developed for agricultural applications. Users can obtain real-time information about the status of the aircraft, the operating system, and other devices connected to the remote control. The application has an integrated intelligent operation planning system. With it, users can intelligently plan plots in the application, and the aircraft will perform operations automatically.

7.1 Main Interface

1. **Aircraft and remote control connection status:** Includes statuses: Aircraft connected, Aircraft not connected.
2. **RTK marker access status:** Displays the number of satellites found by the remote control's GPS or the RTK marker.
3. **Flight battery power:** Displays the remaining flight battery power in real time.
4. **Remote control battery power:** Displays the remaining remote control battery power in real time.
5. **Network connection status:** Displays the network connection status of the SIM card and Wi-Fi, respectively.
6. **Time:** Displays the current time.
7. **Plot management:** Manages the data of plots planned by the locally connected user on the remote control.
8. **Account:** Users can view account information, team information, statistics, and details of operation records.
9. **Device management:** Manages aircraft, remote control, battery, charger, mobile RTK station, RTK module, firmware update, etc.
10. **Settings:** General settings.
11. **Plan plots:** Planning of plot information before route operation. Allows planning of operation plot information, obstacle information, non-operation areas, etc., including aircraft, RTK, and map planning.
12. **Start operation:** Click the "Start" button to access the operation interface, where users can check the aircraft status and perform operations. The default mode is manual operation, which can be switched to different modes, such as route operation, AB operation, assisted manual operation, and manual operation.

7.2 Plot Planning

7.2.1 Add plots

Click the "Plan Plots" button on the main interface to access the operation interface. Below is an explanation on how to add a plot.

1. **Select a planning method and add parcels:** Select **Boundary Point** and the planning method will automatically be set to map planning. Drag the map and click **Add** to add a boundary point at the crosshair's location on the map interface. If users choose RTK planning or aircraft planning, they need to hold the remote control at the corresponding location or control the aircraft to fly to the corresponding location and then click Add. If users need to edit the added points, click to select and drag the points to move them, or click the delete button to erase them.
2. **Mark obstacles:** Select **Obstacle**, drag the map to locate the crosshair at the corresponding position, and click **Add** to mark the obstacle's boundary point. Three or more obstacle boundary points form an obstacle parcel. Select **Circular Obstacle**, click **Add**, and a red circle will appear at the crosshair. After clicking the center of the

- circle to select it, drag to adjust the obstacle's position; after clicking the small red dot on the circumference to select it, drag to adjust the obstacle's radius.
3. **Set up the route:** After planning the parcel, the route will be automatically generated. The green dot on the route indicates the route's starting point, and the yellow dot indicates the route's end point. The following settings can be made for the route.
 - a. **Modify the initial route:** Double-click a boundary line to adjust the route direction to be parallel to the boundary line. Click a point on the boundary line to change the initial route's starting point. Operation line spacing: Set the distance between generated routes.
 - b. **Terrain setback:** Define the distance the route retracts inward relative to the operation area boundary; select Single-sided setback to define the distance the route retracts inward relative to the selected boundary line.
 - c. **Obstacle margin distance:** adjust the distance between the route and the edge of the obstacle.
 - d. **Route division:** Define the route split point to perform route operations in sections.
 4. **Click Save and name the operation route:** After saving, users can view the newly added parcel in the parcel list.

7.2.2 Edit plots

After selecting the plot from the plot list, click **Edit** to edit the plot, which is the same as the plot planning operation. Additionally, the "plot cut" function can also be used to divide the plot into multiple operating areas and set operating parameters separately. After editing is complete, click **Save**.

7.3 Start Operation

Click the **Start** button on the main interface to enter the operation interface.

1. **Return to the home page:** Return to the home page of the GTEEX REVOLUTION Drone app.
2. **Status bar:** Displays the current aircraft status and operation mode during normal flight operations. Warning texts will appear in this column for low battery, low pesticide level, low satellite count, or other real-time statuses.
3. **Operation scenario:** Field or fruit tree scenarios can be switched.
4. **Operation status icon:** Displays the real-time status of distribution or spraying operations, indicating whether they are active or disabled.
5. **Front and rear obstacle avoidance radar status:** Displays the real-time status of the radar, including on, off, not connected, or failed.
6. **RTK status:** Displays the RTK status in real-time, including fixed, single point, and unfixed, displayed in green, yellow, and red, respectively.
7. **RTK base station:** Displays the real-time connection status of the RTK base station, mobile base station, and network RTK, including connected and not connected, displayed in white and gray, respectively.
8. **Remote control connection status:** Displays the wireless connection status between the remote control and the aircraft in real-time and displays the signal strength in the form of a signal grid. Front 1, 2, and 3 represent different joystick modes.
9. **Remote control battery:** Displays the remote control battery level in real-time as a percentage. When the battery level is below 20%, it will be displayed in red numbers to remind users to charge the battery in time.
10. **Network connection status:** Displays the real-time internet connection status and network status of SIM1, SIM2, and WIFI connections, respectively.

11. **Remaining material display:** Displays the remaining material volume or weight in real-time.
12. **Settings:** Click to enter and set parameters for the flight platform, remote control, operating system, radar system, battery, and other systems.
13. **Map operation shortcut buttons:** Includes the map route deletion button, aircraft map centering button, map lock/move toggle button, front and rear FPV interface toggle button, etc.
14. **Operation mode switch button:** The aircraft's operation mode can be switched, including manual operation, assisted manual operation, and AB operation.
15. **Plot list:** Displays a list of planned plots that can be used for operation calls.
16. **One-click takeoff/landing button:** Autonomous takeoff and landing buttons.
17. **Real-time flight and operation data:** Displays the aircraft's speed, altitude, distance, flow rate, and operation area data in real-time.
18. **Image/map toggle button:** Switches between the FPV image interface and the map interface with one click.
19. **Start button:** Click the Start button in manual operation mode to start calculating the operation area.

7.4 Operation Mode

7.4.1 Route Operation

After completing parcel and route planning, users can directly call up parcels to perform route operations. Or users can select and call parcels from the parcel list on the operation interface to perform route operations.

Operational procedures:

1. Place the aircraft near the operating area, users should face the aircraft's tail and turn on the remote control first, then the aircraft.
2. Enter the app's operation interface, click the Parcel List button on the left side of the screen, select the corresponding parcel from the parcel list, and click the Call button.
3. In the operation settings menu, relevant operational parameters can be configured.
4. **Route adjustment:** If there is a deviation between the planned parcel and the actual operating area, users can click **Correct Offset** and adjust the parcel's position via the fine-tune button. Drag the map to add a transfer point where the crosshair is. The transfer point can be used in conjunction with the intelligent start feature to adjust the start route and avoid obstacles that were not marked when planning the parcel
5. Click **Run**, check the aircraft status and operation settings, set appropriate altitude and speed for takeoff/return, and then slide the slider. The aircraft will automatically take off and perform the operation.

 The takeoff/return altitude and speed set here will be synchronized with the parameters in the aircraft settings. If users modify any setting, the other value will be updated simultaneously.

When users click "Run" to automatically take off and start the operation, the aircraft will reach the first waypoint according to the set takeoff altitude. When users click "run" to automatically fly and start operation, the aircraft will reach the first waypoint according to the current altitude.

 *Ensure there is open space near the takeoff point and set appropriate takeoff/return altitudes according to the operating environment.*

During operation, the aircraft will not start operation when flying on transition routes between spray routes and non-operational areas, and will automatically start operation when flying on other routes. Users can adjust the operation amount, flight speed, relative crop height, etc., in real-time on the application interface.

During the operation, the user can pause the operation by moving the roll stick or the pitch stick. The aircraft will hover in place and record the breakpoint. At this point, the user can control the aircraft freely. Then, click the "Continue" button in the lower right corner of the app, and the aircraft will fly to the selected return point and continue the operation. When flying back to the breakpoint, be sure to pay attention to flight safety.

Users can configure the aircraft's action after the operation in the application.

7.4.2 AB Point Operation

Select AB point operation in the operation interface/operation mode. The aircraft can fly and operate along a specific route after marking the points. During operation, efficiency can be adjusted in real-time in the application interface. This mode is suitable for operations in large areas with few obstacles and regular plots.

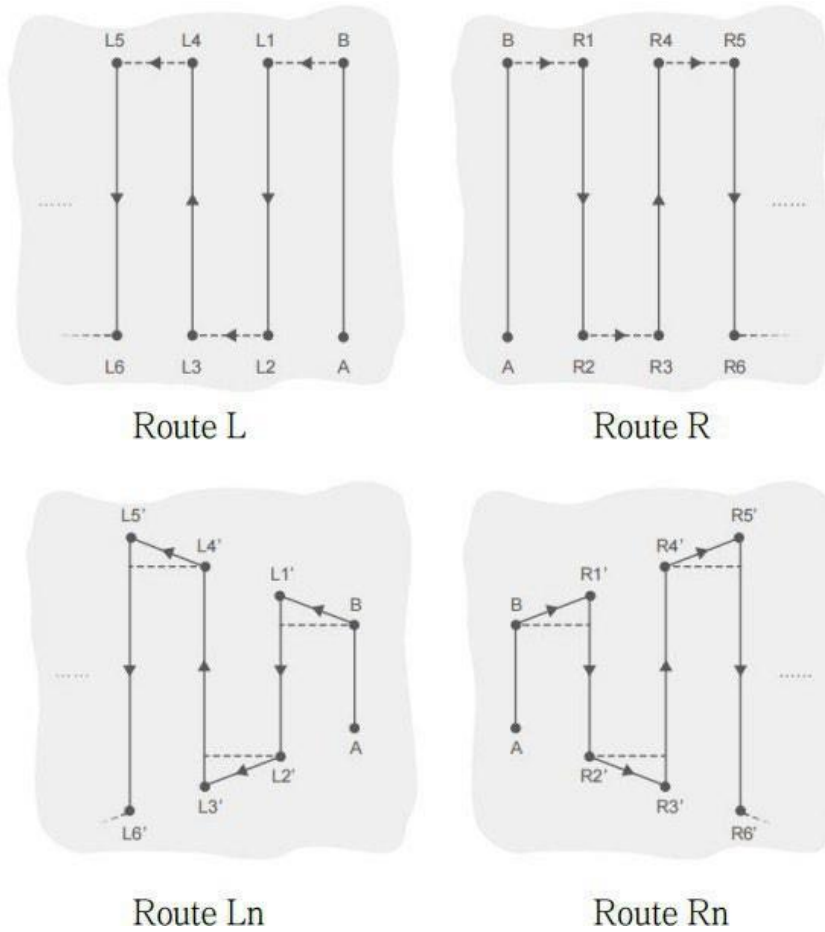
Operational procedures:

1. **Set operation parameters:** Set parameters such as operation amount, flight speed, operation line spacing, relative crop height, etc.
2. **Record points A and B:** Fly the aircraft to point A or B and make it hover. Press button A or B on the application interface, and point A or B will appear on the map, indicating that the recording was successful. If the angle needs to be adjusted, adjust the angle of point A after recording it, then record point B and adjust its angle.

 *While flying from point A to point B, the aircraft will automatically spray pesticides.*

Ensure to record point A first, then point B, and that the distance between points A and B is greater than 16 feet. If there is no pesticide in the spray tank or if the flight speed is greater than 1.25 ft/s, points A and B cannot be recorded. After points A and B are recorded, the position cannot be adjusted. If adjustments are needed, a new AB point operation must be started. During the recording of points A and B, try to ensure that the AB direction is parallel to one of the sides of the operation area polygon to achieve the best operational result.

3. **Adjust the angles of points A and B:** After successfully recording point A or B, click the A or B angle button on the application interface and move the remote control's yaw stick. The aircraft's yaw angle will be the angle of point A or B. A dashed line will be displayed on the map to represent this angle. Click the A or B angle button again to set the current angle as the angle of point A or B.



4. **Select the route direction:** After successfully recording points A and B, the application will generate a route to the right (R or Rn) by default. Click this button to switch the direction and generate a route to the left (L or Ln). During the operation, the user can pause the operation by moving the roll stick or the pitch stick. The aircraft will hover in place and record the breakpoint. At this point, the user can control the aircraft freely. After that, click the "Continue" button in the lower right corner of the application, and the aircraft will fly to the selected return point and continue the operation. When returning to the breakpoint, pay attention to flight safety. Users can configure the aircraft's action after the operation in the application.
5. **Execute operation:** Click the execute button and slide the slider to execute the operation automatically. During operation, ensure the aircraft is within viable range. Also ensure that the GNSS signal in the operating area is good; otherwise, the task will not be completed successfully. If the user uses the pitch or roll stick to control the aircraft during operation, it will automatically switch to manual operation mode and hover after responding to the corresponding joystick action. To continue AB point operation, click "Continue" in the lower right corner of the app, and the aircraft will return to the operation route from its current position. During operation, the aircraft will automatically start operation when flying on a route parallel to the AB line and automatically end operation when flying on other routes.

7.4.3 Manual Operation

This mode is suitable for operations in irregular or small areas. After selecting manual operation on the operation mode search page, the aircraft enters manual operation mode.

In this mode, flow rate and droplet size can be configured, and maximum flight speed can be limited. Users can control the aircraft to the desired area and start operation via the operation button on the remote control. During operation, the flow rate can be adjusted using the buttons on the remote control.

7.4.4 Enhanced Manual Operation

The operation mode button on the interface allows selecting the enhanced manual operation mode. In this mode, operating volume, droplet size, flight speed, relative crop height, and operating line spacing can be configured.

When switching lines, users can use the buttons on the left or right of the interface to make the aircraft automatically fly left or right, respecting the configured operating line spacing. The aircraft automatically starts operation when flying forward or backward and automatically deactivates operation when flying sideways.

If the height adjustment function is enabled, the aircraft's relative height to the crops will remain constant as long as the operating conditions are met.

After activating the "Heading Lock" function, the aircraft's heading will be locked in the current nose direction. Users can control the aircraft to fly in all directions, but the nose orientation will not be controllable. During operation, the operating volume, flight speed, and relative crop height can be adjusted, but the operating line spacing cannot be adjusted.

8. List of Hazards and Risks and Countermeasures

If agricultural drones are not used according to operational requirements or are operated recklessly, hidden risks may arise. Specific hazardous situations and countermeasures are as follows:

Hazardous areas	Responses
Propeller	Maintain a safe distance (more than 100 ft) while the propeller is spinning.
Motor	Maintain a safe distance of more than 1.5 ft when the motor is spinning without propellers installed and more than 100 ft when the motor is spinning with propellers installed.
Pesticide tank	Used pesticide containers contain residues and should not be used to store drinking water or to wash utensils.
Pesticide use	Pesticides must be used according to the manufacturer's instructions or as directed by Plant Protection Department technicians.
During operational flight	It is forbidden to fly in urban areas. Keep away from crowds and maintain a safe distance (more than 100 ft) when operating in the field.
During pesticide spraying operation	Ensure that the operating area is far from bee breeding areas, maintaining a minimum distance of 200 ft, considering prevailing weather conditions.

9. Troubleshooting Instructions

The **GTEEX REVOLUTION Drone APP** issues alarms about failures during use and provides suggestions for resolving them.

9.1 Power System

Device	Problem Description	Suggested Solutions
ESC	The ESC input voltage is too low, and continued flight poses a significant safety risk.	Clean the battery interface and the drone's rear interface with a damp cloth, restart the drone, and check if the issue is resolved. If the issue persists, check if the connectors at the ends of the ESC power adapter cable on the front arm are loose or have water ingress. Replug or clean the connectors. If the cable is corroded or damaged, replace it. If the issue continues, replace the ESC or the drone's rear interface board;
	The ESC temperature is too high.	Cool down and clean the ESC casing, then restart the drone. If the issue persists after cleaning, contact an authorized GTEEX REVOLUTION agent for replacement.
	ESC communication is abnormal, and continued flight poses a significant safety risk.	Check if the connectors at the ends of the ESC communication cable are loose or have water ingress. Replug or clean the connectors. If the ESC communication cable is corroded or damaged, replace it.
Motor	The motor is blocked, and continued flight poses a significant safety risk.	Check if the motor is not spinning smoothly or if the propeller is damaged or deformed. If so, contact an authorized GTEEX REVOLUTION agent for motor or propeller replacement. If there are no visible issues, try restarting the drone.
	The motor's output power is too high, and continued flight poses a significant safety risk.	Check if the motor is not spinning smoothly or if the propellers are damaged or deformed. If so, contact an authorized GTEEX REVOLUTION agent to replace the motor or propellers. If there are no visible issues, try restarting the drone.

9.2 Avionics

Device	Problem Description	Suggested Solutions
IMU	The attitude sensor is abnormal and flight is not permitted.	Try restarting the aircraft or update the firmware to the latest version. If the problem persists, contact an authorized GTEEX REVOLUTION agent to replace the avionics module.
Compass	The compass is abnormal and flight is not permitted.	Try restarting the aircraft or update the firmware to the latest version. If the problem persists, contact an authorized GTEEX REVOLUTION agent to replace the avionics module.
RTK	RTK positioning data is abnormal, caution is recommended for flight.	Try restarting the aircraft or update the firmware to the latest version. If the problem persists, contact an authorized GTEEX REVOLUTION agent to replace the avionics module.
	The secondary RTK antenna signal is abnormal, and the RTK positioning and orientation functions are unavailable.	Place the aircraft in an open and unobstructed environment. Check whether both ends of the RTK feeder cable are properly installed or if there is corrosion. If it is corroded, contact an authorized GTEEX REVOLUTION agent to replace the cable. If the problem persists, move the aircraft to another location to avoid interference. If the problem persists, it may be caused by ionospheric scintillation. Turn off the aircraft and try again after 10 minutes. If the problem still persists, contact an authorized GTEEX REVOLUTION agent. First, try replacing the sub-antenna RTK. If the problem persists, replace the RTK module.
Image Transmission	The image transmission signal is weak and flights are not permitted.	Try restarting the aircraft; Avoid obstructions between the remote control and the aircraft; Raise the remote control antenna; If the problem persists, contact an authorized GTEEX REVOLUTION agent to check the aircraft's image transmission antenna or the remote control.

9.3 Radar System

Device	Problem Description	Suggested Solutions
Radar	Radar connection is abnormal, and obstacle avoidance and altitude adjustment functions fail.	Try restarting the aircraft; Power off the device and check whether the connectors at both ends of the radar signal cable are loose or have water. Reconnect or clean the connectors; If the lower radar cable is corroded or damaged, contact an authorized GTEEX REVOLUTION agent to replace it; If the problem persists, try replacing the radar module or the avionics module.
	The radar temperature is too high and obstacle avoidance performance is unstable.	If the ambient temperature exceeds 100°F, avoid operating in this environment; Clean dirt from the radar surface, wait for radar temperature to drop to normal, and restart the aircraft; If this problem occurs repeatedly, contact an authorized GTEEX REVOLUTION agent to replace the radar module.
	The radar communication is interfered, and the functions of obstacle avoidance and altitude adjustment fail.	Turn off the phone and check if there is any communication interference around; Try replacing the radar module.

9.4 Spraying System

Device	Problem Description	Suggested Solutions
Spraying System	The pump motor self-test is abnormal and cannot operate.	Try restarting the aircraft; Check whether the battery is installed correctly; Check whether the connectors at both ends of the water pump signal cable are loose or have water ingress; If the water pump signal cable is corroded or damaged, replace the cable; If the problem persists, contact an authorized GTEEX REVOLUTION agent to replace the pump module.
	The flow meter connection is abnormal and the accuracy of the spray flow control is low.	Check whether the connectors at both ends of the flow meter connection cable are loose or if water has entered, and reconnect or clean the connectors; If the flow meter connection cable is corroded or damaged, contact a GTEEX REVOLUTION agent to replace the cable; If the problem persists, replace the flow meter.

10. Storage, Transport, and Maintenance

10.1 Storage and Transport

⚠ If it is necessary to transport the aircraft, ensure that the battery has been removed from the aircraft, fold the propellers and secure them in the storage clips on the sides of the aircraft before transport. If long-term storage or long-distance transport is required, remove the battery and spray tank from the aircraft and ensure there is no liquid in the spraying system.

Storage & Transportation Safety Protocol

To prevent injury and property damage:

1. Keep all components away from children – small parts and cables pose choking hazards.
2. Always remove batteries from the aircraft before transport.
3. For long-term storage or extended transport: Detach the spray tank assembly or completely drain residual liquid. Store aircraft in climate-controlled environments.
4. For long-term storage, maintain batteries at approximately 60% state of charge .

The GTEEX REVOLUTION Drone utilizes two GTR 101 Intelligent Batteries. Each GTR 101 battery has a capacity of 30,000mAh and a voltage of 67.5V.

The GTR 101 Intelligent Battery must only be charged using the Intelligent Charger provided with the kit.

When charging with public power sources, use a cable harness rated for a maximum of 15A. Maximum charging power is 3000W. To protect the safety of the power supply input, always set a safe output power level on the charger panel.

1. **Carrying Handle:** Use the handle to lift the battery. After extended use, always check that the handle is secure and reliable before lifting.
2. **Power Button:** Press and release to check battery level. Press and release, then press and hold to turn the battery on. Always ensure the battery is fully inserted into the aircraft before powering it on. Always power the battery off before removing it from the aircraft. Failure to do so may damage the drone and battery interfaces.
3. **LED Indicators:** Display battery level and assist in quickly identifying fault causes (troubleshooting).
4. **Power Port:** Connects to the aircraft or the intelligent charger.
5. **Silicone Protective Sleeve:** Provides shock absorption and protects the battery.

10.2 Maintenance

To ensure the product is in its best condition and reduce safety risks, it is recommended to regularly check and perform maintenance on the aircraft's parts. Please visit the official website to read the "Disclaimer and Safe Operation Guidelines" and the "Maintenance Manual and Warranty Terms" for information on product cleaning and maintenance operations.

Cleaning Preparation: Allow the drone to cool to ambient temperature before cleaning. Never clean immediately after operation. Clean drone and remote controller daily after flight operations.

Step-by-Step Cleaning Procedure:

1. Tank Flushing: Triple-rinse the liquid tank. Fill with clean water or soap solution. Spray until empty and repeat this process two additional times.
2. Remove and clean the tank filter, nozzle screens, and spray tips. After confirming there are no blockages, soak these components in clean water.
3. Rinse airframe with low-pressure water, scrub with soft brush or damp cloth and dry thoroughly with lint-free cloth.
4. For motors, propellers, and heat sinks. Wipe pesticide residue or dust with damp cloth. Immediately dry with absorbent cloth. Never allow liquid pooling.

10.3 GTR 101 Flight Battery Use and Storage Protocol

For long-term storage, charge to 70% before storage (optimal preservation state). Check remaining charge monthly. If the state of charge (SOC) falls below 40%, recharge to approximately 70% before storage. Prolonged low-charge storage causes permanent capacity degradation and reduces cycle life.

1. After connecting the battery to the aircraft, power on: Short-press, then long-press the power button.
2. Power off (after landing): Short-press, then long-press to shut down. Disconnect from the aircraft.
3. Ensure battery level exceeds 90% before each flight.
4. Low-battery alert: Land immediately and replace the battery.
5. Cold weather operation: Pre-warm batteries above 45°F; 68°F recommended. Achieve this by hovering briefly.

Critical Precautions:

1. Prohibited: Use near heat sources (do not leave in direct sunlight, hot vehicles, flames, heaters, or generator exhaust).
2. Never expose to liquids. Water contact may cause thermal runaway, fire, or explosion. Avoid rain/humid environments.
3. Do not use swollen, leaking, or damaged batteries. Contact your dealer immediately.

4. Always power off before installing/removing batteries. No Hot-swapping batteries.
5. Use the battery within an ambient temperature range of 41°F to 113°F. Excessive heat (above 122°F) may cause fire or explosion. Extreme cold (below 40°F) severely reduces performance; normal function resumes at room temperature.
6. Avoid strong electrostatic or magnetic fields, which may trigger protection circuit faults.
7. Never disassemble the batteries, only authorized locations can safely work on internal battery components.
8. Discard batteries after crashes, falls greater than 36 inches or drops.
9. Water immersion, send for inspection. Do not reuse.
10. Fire response: In case of a Fire immediately call 911 and notify the proper authorities.
11. Never short-circuit terminals with metal objects.
12. Avoid impacts or compression. No heavy objects on batteries or chargers.
13. Clean terminals with dry cloth to prevent poor contact or charging failure.
14. Land immediately if battery falls below 15% , as this may damage the battery or cause flight accidents.
15. Reverse polarity PROHIBITED. Improper charging may cause overheating, explosion, or fire. Only use GTEEX REVOLUTION Batteries and authorized Charging stations. Unauthorized batteries may lead to accidents or malfunctions, for which the user is responsible. Unauthorized batteries void warranty; user assumes all liability.
16. Place on flat surfaces to avoid puncture by sharp objects.
17. Danger: Never stack items on batteries, as this may cause damage or danger.

To prevent injury and equipment damage:

1. After daily operations, allow aircraft to reach ambient temperature before cleaning. Never clean immediately post-flight.
 - a. Fill with clean water. Spray until fully emptied and repeat twice.
 - b. Removing and cleaning: Spray tank filter assembly, Nozzle screen filters, Spray tips. Verifying all parts are obstruction-free.
 - c. The entire unit can be washed directly with water. It is recommended to rinse the body using a spray hose (no pressure washing), then clean it with a soft brush or damp cloth, and finally wipe it dry with a clean, dry cloth.
 - d. If there is dust or chemical residue on the motor, propeller blades, or heat sink surfaces, it is recommended to clean them with a damp cloth and then wipe dry with a clean, dry cloth.
 - e. Keep the aircraft in a dry location.

2. After each day of operation, wipe the remote controller's surface and screen with a clean, damp cloth (wring out excess water).
3. After every day of flying or 100 completed Flights:
 - a. Inspect the propellers for cracks. Replace any cracked propeller.
 - b. Check if the propellers are loose. If looseness is found, replace the propeller and washers.
 - c. Examine plastic and rubber components for signs of aging/deterioration.
 - d. Check the nozzle's spray pattern. If atomization is poor, thoroughly clean the nozzle or replace it.
 - e. Check and Clean if necessary both the nozzle filter and the spray tank filter.
4. After daily operations: If the equipment will not be used again the next few days, perform slow charging on the batteries overnight for maintenance.
5. Do not attempt unauthorized repairs on the aircraft. If damage occurs, contact an authorized dealer for service.
6. To prepare the aircraft for transport, remove the battery from the drone. Fold the propellers and fully retract the arms, securing them into the storage clips on the side of the aircraft. Store the aircraft in an way to protect it from the outside environment
7. For long-term storage or long-distance transport, always remove both the battery and the spray tank system from the aircraft. Thoroughly flush the spraying system with clean water to ensure all traces of chemicals are removed. Allow the system to dry completely before repacking. Only transport the equipment after it has been fully dried and secured.

Note:

1. Keep the Lidar module's protective cover clean. Gently wipe the surface with a soft, damp cloth and allow it to air-dry.
2. Maintain FPV camera cleanliness. First, remove any dust, sand, or debris from the camera surface.
3. Inspect all aircraft components for signs of severe impact. If any damage is suspected, contact customer support or an authorized dealer.

Flight Kit (I19K2)	Quantity
INDEPENDENCE 19 Drone with Liquid Tank	1
GT402 Remote	1
GTR101 Intelligent Battery	3
GTR Intelligent Charger	1
Solid Tank Spreader and Landing Gear	1
I-19 Drone Operator's Manual	1

Parameter	Specification
Model	INDEPENDENCE-19
Structure	Four-axis
Dimensions	130x141x33in (arms and propellers unfolded) 50in × 34in × 34in (arms folded)
Max wheelbase	103in
Weight	152lb (with battery) 110lb (without battery)
Tank capacity	19 gal
Hovering flight accuracy	RTK activation: ±4 in (horizontal), ±4 in (vertical)
Max configurable flight radius	1.2 miles
Hovering flight time without load	16min
Hovering flight time with max load	6min

Power System	
Motor KV value	46 (r/min)•V
Motor size	5.5x1.4in
Motor rated power	4650W x 4
Rated speed	2000r/min
Main rotor diameter	62 inches
Number of main rotors	4
Main rotor material	Carbon fiber

Spraying System	
Spray tank capacity	19 gal
Spray type	Centrifugal
Total number of nozzles	2/4 centrifugal nozzles
Max spray width	38 ft (operating height: 11.5ft± 1.6ft, flight speed: +33ft/s)
Atomized particle size	Centrifugal nozzles: 50 μm - 500 μm
Pump type	Magnetic Impeller Pump
Number of Pumps	2
Max Flow	5gal/min (two centrifugals) 8gal/min (four centrifugals)

Remote Control System	
Model	KG402
Image Transmission	Dual band 2.4/5.8G SRRC: \>1.9mi CE: \>1.9mi FCC: \>1.6mi (in open space without interference, aircraft height 1.6mi)
RTK Module	GNSS: GPS/BeiDou/GLONASS Positioning accuracy: 0.3mi ± 1ppm (RTK)
Monitor	6-inch touchscreen, 1280x720 resolution
External batteries	2
Total battery life	≥ 8 hours
Charging power	External: 33.6W/8.4V/4A x 2

Camera System	
Front	FPV Camera
Max resolution	1920x1080

Radar System	
Fixed height range	Measurement height: 3 - 180 ft Height range
Obstacle avoidance range	Obstacle avoidance distance (horizontal): 3.3-164ft Field of view (3db): horizontal 50°, vertical 10° Usage conditions: Aircraft's relative height must be greater than 6.6ft and speed less than 23ft/s Safety distance: 8.2ft (distance between propeller tip and obstacle after braking and aircraft stabilization)

Note: When flight speed is less than 16.4ft/s and spray width is less than 22ft, the minimum pesticide application rate is 4 gal/acre. When flight speed exceeds 16ft/s and spray width exceeds 26ft, the minimum pesticide application rate for this product is 4 gal/acre, otherwise, this will affect the drone's operational autonomy.

GTEEX REVOLUTION Manual Disclaimer

1. Thoroughly review this manual before product use. By activating this product, you are deemed to have acknowledged, understood, accepted, and agreed to all terms and conditions contained herein.
2. Agricultural drones carry operational risks. GTEEX REVOLUTION shall bear no liability whatsoever for any consequences arising from drone operation by minors or Unauthorized users. Under the terms of the purchase of this Drone, all users shall complete their drone startup check list at activation or the manufactures warranty shall be voided after 6 flights without completing it. Flights by unauthorized users will void the warranty.
3. Users are strictly prohibited from operating the drone under the following conditions: While under the influence of alcohol, drugs, or medication. While experiencing dizziness, fatigue, nausea, or any other physical or mental impairment. GTEEX REVOLUTION disclaims all liability for any incidents or damages resulting from such prohibited operation.
4. When handling toxic or restricted pesticides: Thoroughly adhere to the pesticide manufacturer's labeling. Implement all specified personal protective equipment (PPE). GTEEX REVOLUTION shall bear no liability under any circumstances for injuries to persons, animals, or plants resulting from user negligence or improper chemical application.
5. This product is a multi-rotor agricultural drone designed exclusively for spraying, spreading, and cargo carrying applications within agricultural, forestry, livestock, and aquaculture operations. Any use beyond this specified scope is strictly prohibited. GTEEX REVOLUTION shall bear no direct, indirect, or consequential liability whatsoever for incidents resulting from non-compliant operations.
6. Unauthorized modification or use of non-GTEEX REVOLUTION components is strictly prohibited. GTEEX REVOLUTION shall bear no direct, indirect, or consequential liability for any incidents, damages, or injuries resulting from such unauthorized alterations or use of third-party parts.
7. To optimize product functionality and enhance user experience, you expressly acknowledge, understand, and consent that during product operation, all flight records and data will be automatically uploaded and stored to GTEEX REVOLUTION's secure server. GTEEX REVOLUTION may lawfully collect, store, and utilize all related operational data generated during your use of this product. Should flight data fail to be uploaded due to your actions or omissions, GTEEX REVOLUTION shall bear no liability whatsoever for: Any impacts on flight safety, compromised product or service quality, operational consequences arising from unavailable data for storage and analysis.
8. To the maximum extent permitted by law, GTEEX REVOLUTION, its distributors and Dealers shall not be liable for any losses resulting from the failure to operate the product in accordance with the "Operator's Manual" nor for any indirect, consequential, punitive, incidental, special, or exemplary damages, including but not limited to damages arising from your purchase, use, or inability to use the product.
9. You acknowledge and agree that during product use, unforeseen incidents may occur

due to operational errors, environmental factors, network communication issues, or a combination thereof. These incidents are considered inherent and reasonably foreseeable risks of product operation, and GTEEX REVOLUTION, its distributors and dealers shall not be held liable for any such occurrences.

10. To the maximum extent permitted by law, in no event shall GTEEX REVOLUTION liability to you for all damages, losses, and causes of action (whether in contract, tort, or otherwise) exceed the amount you actually paid to GTEEX REVOLUTION, its distributors or its authorized dealers for the product.
11. Under all circumstances, purchasers or users must comply with applicable laws, regulations, and policies of the country or region where the product is operated. GTEEX REVOLUTION, its Distributors, and dealers shall bear no liability whatsoever for violations of such legal requirements by purchasers or users.
12. Certain jurisdictions may legally prohibit disclaimers of liability. Consequently, your rights may vary across different states. This shall not be construed as automatic invalidation of any provisions herein where legally permissible, the laws governing this manual are the state of North Carolina and you agree that should any claim arise, it will be in the Jurisdiction of the state of North Carolina in Catawba County.
13. To the maximum extent permitted by law: GTEEX REVOLUTION reserves the sole right to interpret and modify these terms. GTEEX REVOLUTION reserves the right to update, modify, or terminate the terms of the " User Operation Manual " at any time without prior notice, through official channels including but not limited to the GTEEX REVOLUTION website and App, you as the customer can request any updated versions by contacting the manufacturer at any time and an updated copy will be provided to you.

FCC Statement

Any Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) This device may not cause harmful interference, and
- (2) This device must accept any interference received, including interference that may cause undesired operation

FCC Radiation Exposure Statement for drone:

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 7 in between the radiator & your body.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

RF Exposure Statement for remote controller:

This device meets the government's requirements for exposure to radio waves. This device is designed and manufactured not to exceed the emission limits for exposure to radio frequency (RF) energy set by the Federal Communications Commission of the U.S. Government.

The exposure standard for wireless devices employs a unit of measurement known as the Specific Absorption Rate, or SAR. The SAR limit set by the FCC is 1.6 W/kg. *Tests for SAR are conducted using standard operating positions accepted by the FCC with the device transmitting at its highest certified power level in all tested frequency bands.

Although the SAR is determined at the highest certified power level, the actual SAR level of the device while operating can be well below the maximum value. This is because the device is designed to operate at multiple power levels so as to use only the power required

to reach the network. In general, the closer you are to a wireless base station antenna, the lower the power output.

For body worn operation, this device has been tested and meets the FCC RF exposure guidelines for use with an accessory that contains no metal and the positions of the device a minimum of 0mm from the body. Use of other enhancements may not ensure compliance with FCC RF exposure guidelines. If you do not use a body-worn accessory, position of the device a minimum of 0mm from your body when the device is switched on at its highest certified power level in all tested frequency bands.

The FCC has granted an Equipment Authorization for this device with all reported SAR levels evaluated as in compliance with the FCC RF exposure guidelines. SAR information on this device is on file with the FCC and can be found under the Display Grant section of <http://www.fcc.gov/oet/fccid> after searching on FCC ID: 2BBNT-HD402.